Global CO_2 Emissions: Trends, Regional Shifts, and Per Capita Inequalities

A data-driven overview from the Industrial Revolution to 2024

Sources: Global Carbon Budget 2025 · NASA Climate Vital Signs · Our World in Data

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

Latest measurement from NOAA's Mauna Loa Observatory (February 2026). More than 50% above the pre-industrial baseline of ~280 ppm.

- Current level is at least 1.5× greater than the natural increase at the end of the last ice age (20,000 years ago)
- That ancient change took thousands of years; the modern rise has occurred in roughly two centuries
- Ground-based and satellite measurements confirm sharp acceleration linked primarily to burning coal, oil, and natural gas

Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

1750

Negligible

Pre-industrial baseline

1950

~6 Bt

Post-war industrial acceleration

1990

~20 Bt

Nearly quadrupled in 40 years

2024

>35 Bt

Growth slowed, but not yet peaked

Figure: Annual CO_2 emissions by country (figure_3.jpg)

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Country / Region	Approx. Annual CO_2 (billion t)	Share of Global	Trend (recent)
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
Brazil	~0.5	~1.4%	Roughly stable
France	~0.3	~0.8%	Declining

Figure: Annual CO_2 emissions by country (figure_3.jpg)

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900

Europe + US

>90%

Industrialized West dominated global emissions. Asia, Africa, and Latin America contributed marginal shares.

2024

Asia

~50%

Asia produces roughly half of all emissions. US + Europe combined account for <33%. Africa and South America each contribute only 3–4%.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

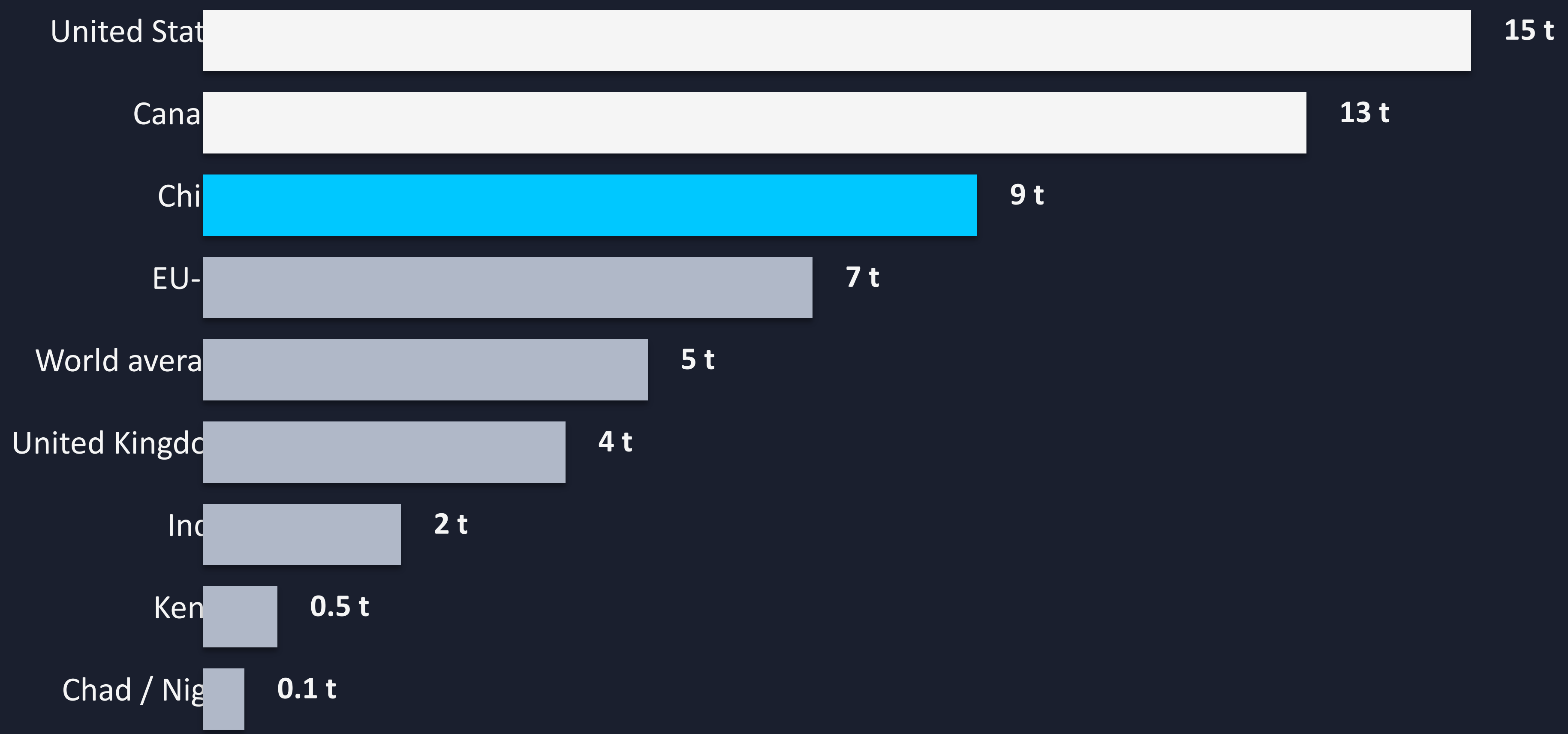
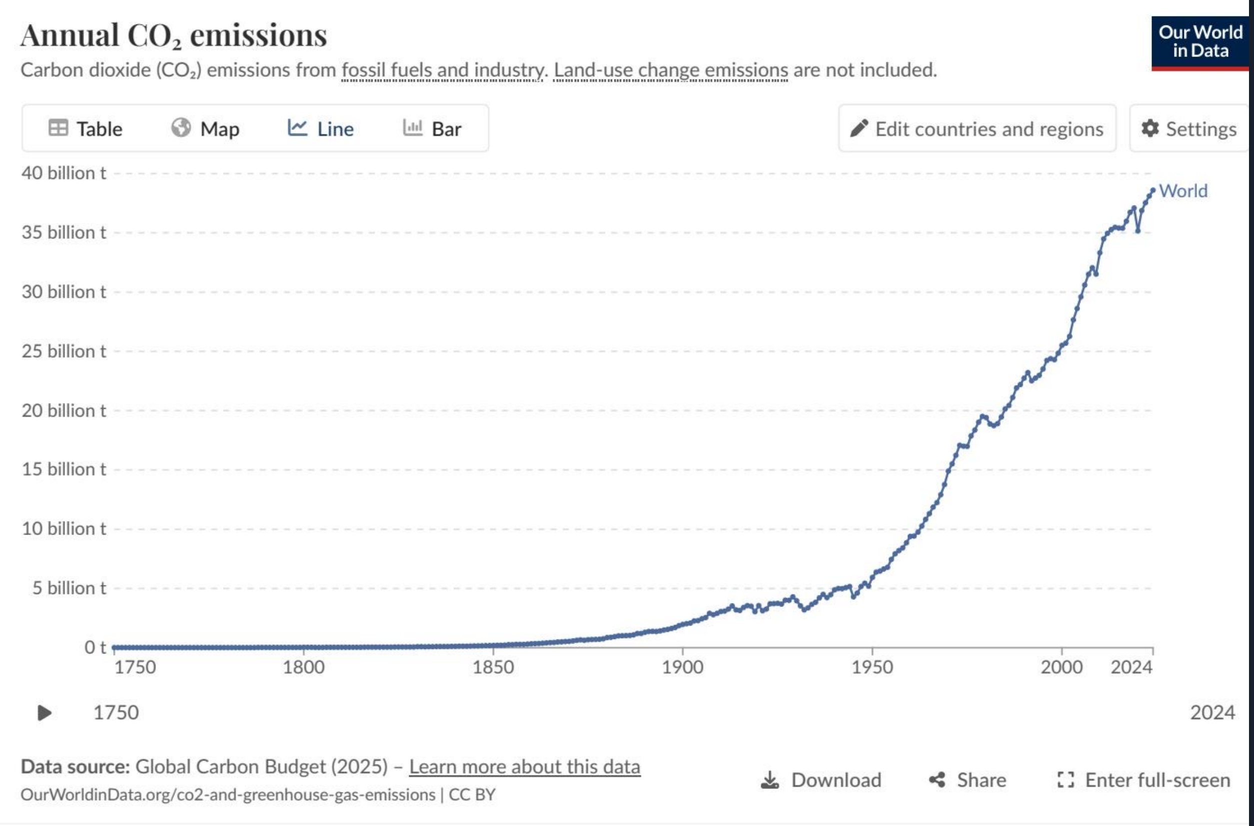


Figure: Per capita CO_2 emissions over time (figure_1.jpg)

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase



Declines (blue):

• United States and much of Europe: –1% to –5%

Increases (orange/red):

• Russia, parts of Southeast Asia, and several African nations: +2% to +10%

• Some South American countries: >+10%

Interpretation:

• The global map of emissions growth is fragmenting.

• Mature economies are decoupling; emerging economies are still climbing.

• Percentage changes can mask small absolute bases in low-emitting nations.

Figure: Annual percentage change in CO_2 emissions, 2024 (figure_4.jpg)

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Metric	France / UK	Germany / Netherlands
Primary energy mix	High nuclear + renewables	Higher fossil fuel share
Per capita CO_2 (approx.)	~4 tonnes/year	Higher than France/UK
Nuclear share	Significant (France)	Phasing down (Germany)
Policy implication	Decarbonization via low-carbon baseload	Transition complexity with coal/gas reliance

Key insight: Countries with similar prosperity levels can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels. Policy and infrastructure choices matter as much as GDP.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1. Global emissions exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.

2. China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is the third-largest and rising fast.

3. The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today, while the US and Europe combined are below one-third.

4. Per capita inequalities remain extreme — a 150× gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person).

5. Energy mix and policy choices matter: countries with similar prosperity levels (e.g., France vs. Germany) can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels.